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PERMIAN PALEOMAGNETISM OF THE NORTHERN TIEN SHAN AND ITS TECTONIC IMPLICATIONS

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Abstract. As a part of structural and paleomagnetic investigations of the late Paleozoic and younger tectonism in the Tien Shan, Permian sedimentary and volcanic rocks were sampled at four localities within the Northern Tien Shan west of the Issyk Kyl Lake. After complete thermal demagnetization, prefolding characteristic components were isolated for four localities, one of which also yielded a synfolding magnetization. When compared with the European reference data, the inclinations of these Permian rocks appear to be close to the reference ones. Mean declination for the northernmost locality also agrees with the European data, while the results for the southern parts of the area point to counterclockwise rotations. Comparison with data from an independent study (Thomas et al., 1992) shows that the rotations predate Alpine deformation but probably postdate the initial stage of the folding which affected the Permian rocks. Analysis of paleomagnetic data from different parts of Kazakhstan together with our results from the Northern Tien Shan has led to the conclusion that these blocks had already been welded with the European plate in Permian time.

INTRODUCTION

Paleomagnetic investigations are useful to test whether a block or a terrane has always been in its current position and orientation in a fold belt. This method is most successful for unravelling tectonic evolution if paleomagnetic data are numerous and their deviations from any reference frame are due to the latest deformations in an area (e.g., Southern California [Luyendyk, 1989]). Tectonic interpretation of paleomagnetic results is much more difficult if the data come from rocks which suffered a multiphase deformation and later movements are not well understood. However, if later movements can be progressively accounted for with the aid of structural and paleomagnetic data, a consistent pattern might emerge, and the directions for the older rocks will become more useful.

The Tien Shan is a mobile belt affected by multistage deformation. For instance, four periods of folding and thrusting occurred in the South Tien Shan during the Carboniferous-Permian [Burtman, 1976]; in addition, this region was affected by Alpine (late Cenozoic) deformation. Many would like to understand the lower and middle Paleozoic evolution of the fold belt. Some Silurian-Devonian paleomagnetic results may be of great help here [e.g., Didenko et al., 1988], but their interpretation is hindered by the scarcity and often low quality of the paleomagnetic data on late Paleozoic and younger rocks. The current research program is directed toward structural and paleomagnetic investigation of the late Paleozoic and Alpine tectonics of Central Asia. Below are reported the results on Permian rocks from the Northern Tien Shan.

GEOLOGICAL SETTING AND SAMPLING

The Tien Shan is usually divided into three approximately E-W trending tectonic realms (Figure 1). Separated from zones in the south by the Nikolaev Line fault, the Northern Tien Shan fold belt comprises Riphean to lower Paleozoic rocks and is usually regarded as the southernmost part of the Kazakhstan Caledonian realm [Khain, 1977]. The Caledonian complex is unconformably overlain by Devonian volcanics and Carboniferous red beds. Occasional volcano-sedimentary rocks represent the uppermost member of the Paleozoic succession and, judging by floral remains and crosscutting rela-

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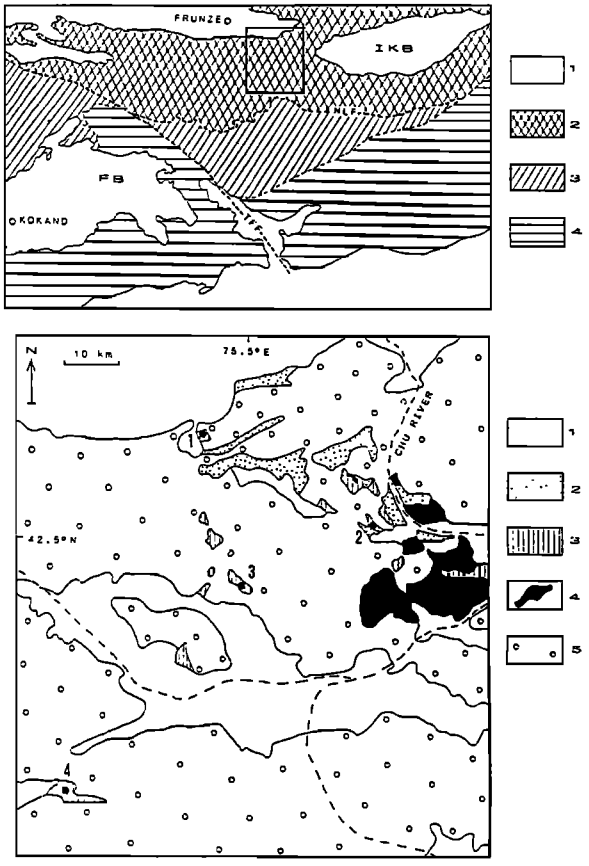


Fig. 1. Schematic tectonic zonation of the Tien Shan (top). Legend: 1, Mesozoic and Cenozoic cover (FB, Fergana basin; IKB, Issyk-Kul basin); 2, Northern Tien Shan; 3; Central Tien Shan; 4, Southern Tien Shan. Main faults are shown as dashed lines (TFF, Talas-Fergana fault; NLF, Nikolaev Line fault). Simplified geological map of the studied area (bottom). Legend: 1, Pliocene-Quarternary cover; 2, Paleogene-Miocene rocks; 3, Permian volcanics and volcano-sedimentary rocks; 4, Permian intrusions; 5, Pre-Permian rocks. Sampling localities (solid dots) are numbered as in the text.

tions with granites, some of which have K-Ar whole-rock ages, these rocks are of the Late Carboniferous-Permian, more likely Permian age [Masumov et al., 1978]. These subaerial volcanic sequences had probably been more widespread but were later destroyed by erosion.

In general, the Tien Shan is regarded as a part of the Eurasian plate since the Permian and little is said about its late Paleozoic tectonics; only simple folding and subvertical faulting have been reported [Khain, 1977]. An exception is the region just south of the large Talas-Fergana fault, TFF (Figure 1), where arcuate structures and intense folding related to dextral strike slip along this fault have been described [Burtman, 1976].

Triassic rocks are almost absent from the Tien Shan. Jurassic coal-bearing deposits are confined to small basins and overlie Paleozoic rocks with a major angular unconformity. Thus the final stages of the Variscan movement can be dated as Permian-Triassic, more probably as Late Permian-Early Triassic. To the southwest of the TFF (Figure 1), continuous sedimentation took place from the Early Cretaceous to the Miocene, while to the northeast, subaerial clastic sedimentation began in the mid-Paleogene. Alpine deformation probably started in the earliest Neogene and reached a maximum in the Pliocene [Khain, 1977].

In order to investigate late Paleozoic deformation far away from TFF, Permian rocks were sampled at four localities west of Issyk-Kul lake (Figure 1). Permian volcanics, exposed as outliers, are often intruded and reworked by Permian granites, especially in the eastern part of the area. As the temporal relationship between this magmatic activity and the folding of stratified rocks is not clear, the sampling localities were chosen at some distance from granitic bodies in order to diminish the possibility of a synfolding or postfolding remagnetization.

Volcanoclastics at the northern slopes of the Kirghiz Ridge generally dip northward, but both strikes and dip angles vary (locality 1). A southeastward dipping lava sequence was sampled at the eastern termination of the same ridge (locality 2). On the southern slopes of the same ridge, sediments rich in volcanoclastics and lava flows were studied across a syncline with gentle northern and steep southern limbs (locality 3). At the same locality, 11 boulders were sampled from intraformational conglomerate, consisting of volcanoclastics with well-rounded lava boulders up to 15 cm in diameter.

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Both limbs of a large syncline on the northern slope of the Moldotau Ridge were studied (locality 4). The succession consists of andesites (300 m), tuffaceous rocks (200 m) and red sandstones (100 m). This rocks are Permian on the basis of pollen data [Osmonbetov, 1982]. Andesites at the steep southern limb are intruded by a gabbro body, subparallel to bedding, whereas almost flat-lying red beds at the other limb are cut by a number of vertical diabase dykes. The intrusives are reportedly Permian [Osmonbetov, 1982]. Both limbs are monoclines, but the central part of the fold is rather severely deformed; some layers here are overturned. Samples were taken from both stratified rocks with different attitudes and intrusive bodies. In order to perform a contact test, three further samples were taken from a single red bed layer at distances of 0.1, 0.5, and 1.5 m from a dyke about 0.5 m thick.

Altogether, 118 hand samples oriented with a magnetic compass were taken from four localities. When possible, samples were taken along long continuous sections, a sample per a layer of sediment, lava flow, or dyke. Sampling points were spaced every few meters to several tens of meters, and the total true thicknesses studied were up to several hundred meters. Due to poor exposure, no thick sections were found at locality 4, and samples were taken from several thin ones with thicknesses of less than 50 m.

METHODS AND PROCEDURES

Representative pilot samples (about 25% of the total) were subjected to detailed stepwise thermal demagnetization. According to the results thus obtained, the optimal steps were chosen for the remaining part of the collection. Two specimens from a hand sample were always heated, their mutual orientation having been changed after each step. All measurements were made on a Czechoslovakian spin magnetometer (JR-4). The data were analyzed with stereonets and orthogonal diagrams. After isolation of components, mean vectors were calculated for each hand sample. These sample means, being independent spot readings of the ancient geomagnetic field, were in turn used for computation of locality means or site means.

The fold test, applied to each locality separately, was performed in two ways: as a test (CP test) based on the ratio of in situ and tilt-corrected concentration parameters [McElhinny, 1964] and as another one (M test) based on comparison of mean vectors [McFadden and Jones, 1981]. For the latter, sample means were used instead of site means, and the distribution of normals to bedding planes for all sampling points was divided into a number of groups, irrespective of sampling point position within a locality [Bazhenov and Shipunov, 1991].

RESULTS AND COMPONENT DATING

NRM intensities in the collection ranged from 5000 nT (magmatic rocks) to about 10 nT (red beds). An unstable part of magnetization, presumably of viscous origin, was destroyed after the first heating to 200°C; magnetization intensities dropped about 20% on average. The unstable component yielded no usable information and was excluded from data presentation and subsequent analysis. With few exceptions, all samples yielded consistent results, magnetization intensities even at the final steps of heating being at least 10 times larger than the noise level of the magnetometer.

Volcano-sedimentary rocks from locality 1 responded well to thermal cleaning. The characteristic magnetization (ChRM) of reversed polarity was isolated in 19 out 22 samples (Figure 2a). ChRM directions are tightly clustered (Figure 3a) and the locality mean is well defined (Table 1). This remanence passes both modifications of the fold test (Table 2).

Volcanics from locality 2 showed univectorial decay to the origin above 200°C. ChRM directions are well grouped (Table 1). Bedding attitudes are uniform here, and the fold test could not be performed.

A ChRM was easily isolated in volcanics and sediments from locality 3 (Figure 2b). The ChRM directions for two specimens from a sample agreed well, but sample means were rather scattered (Figure 3b). The scatter is likely due to structural reasons,

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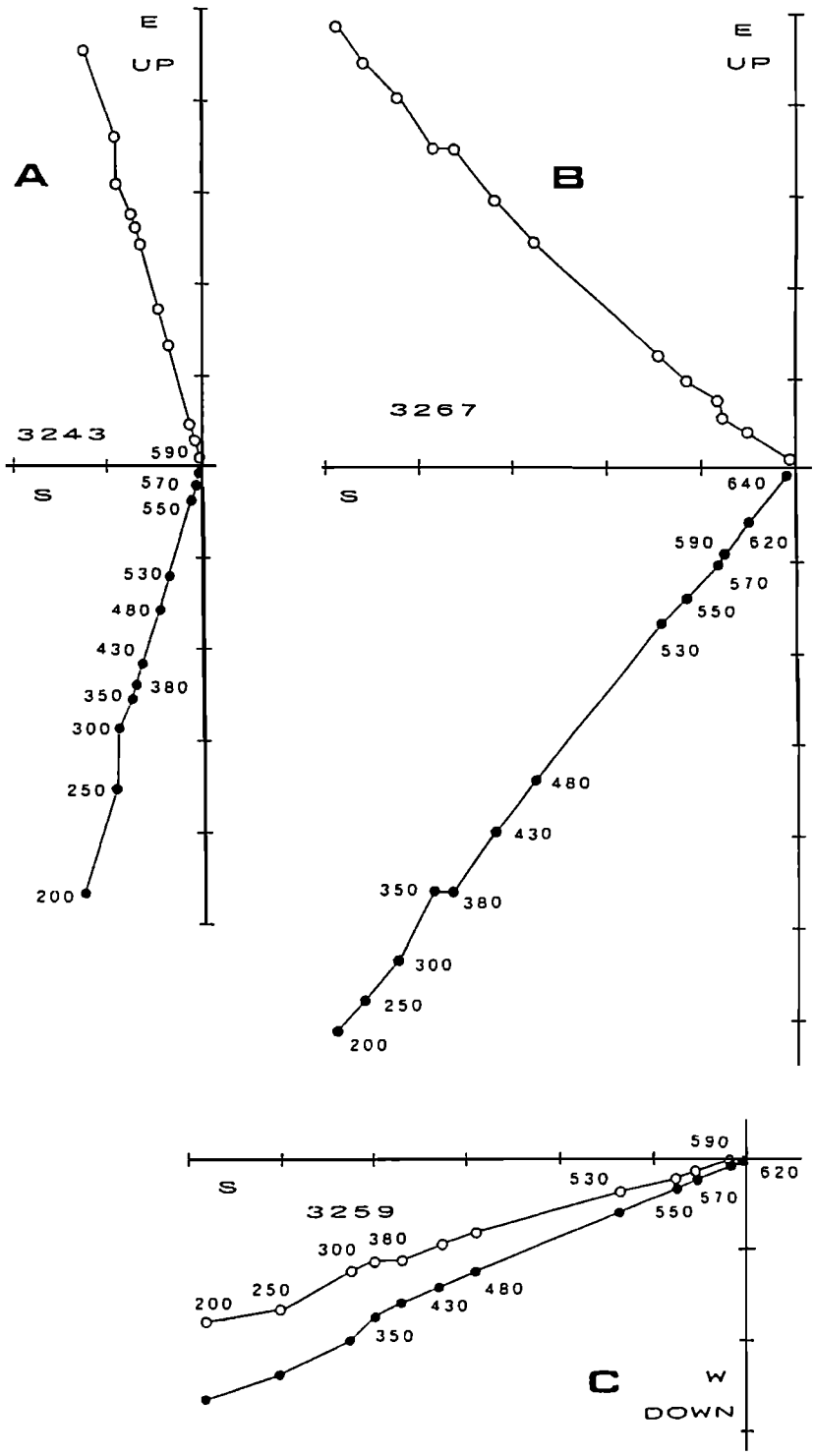


Fig. 2. Representative vector plots of (a) volcano-sedimentary rocks from locality 1, (b) lava boulders, and (c) lava flows from locality 3. Directions are plotted in sample coordinates. Solid (open) dots represent vector end-points projected onto the horizontal (north-south vertical) plane. Magnetization units on axes are arbitrary; temperatures are given in degrees Celsius.

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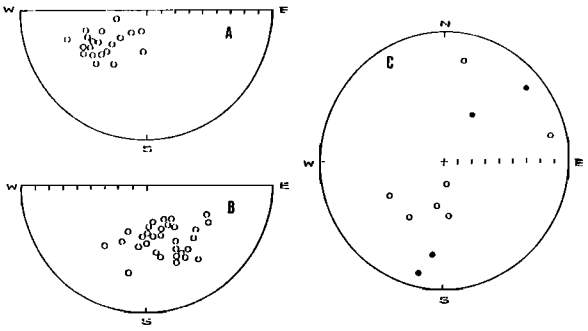


Fig. 3. Sample mean directions of characteristic component (ChRM) after tilt corrections from (a) locality 1, (b) locality 3A (localities 2 and 3 combined), and (c) lava boulders from intraformational conglomerate. Open (solid) symbols are plotted onto the upper (lower) hemisphere of equal-area projection.

as it was difficult to measure bedding precisely, especially for volcanics; the readings below and above a lava flow often differed by 15°-20°. This source of dispersion is probably random. It has not led to any bias, as the mean vectors from the fold limbs agree excellently after tilt correction: D = 160°, I = -40° and D = 160°, I = -40° for 12 and 10 sample means from the northern and southern fold limbs, respectively. The data from localities 2 and 3 were combined and treated together (further on referred to as locality 3A). The positive fold test suggests a prefolding age of the ChRM here (Table 2).

A ChRM was isolated in lava boulders (Figure 2c), but the conglomerate test yielded controversial results. According to the test of uniformity [Mardia, 1972], the calculated vector-resultant is 0.33, whereas the critical value for 11 unit vectors is 0.48; thus, remanence directions from lava boulders are random. However, the distribution on the stereonet is clearly not chaotic, and a cluster of five reversed directions in the SW sector, rather similar to the data for this locality, can be recognized (Figures 3b and 3c). Boulders were taken from three conglomerate layers, each about 10 m thick, intercalated with lava flows of similar or larger thicknesses. Some boulders might have been remagnetized during lava emplacement. It would be interesting to compare remanence directions in boulders versus distance from the overlying flow, but unfortunately such data were not recorded in the field. Nevertheless, the general scatter of paleomagnetic vectors from these boulders points to the primary nature of their ChRM.

At locality 4, the component separation was not a problem with volcanics (sites 1 and 2 and dykes from site 9). Most samples showed univectorial decay to the origin above 200° or 250°C (Figures 4a and 4b). It was not always so with sediments. At some sites (4, 6, 7, and red beds from site 9), a univectorial component was isolated above 300°C. In contrast, sample directions in violet-brown tuffaceous rocks (site 3) and red beds (site 8) shifted along great circles during cleaning, suggesting a multicomponent magnetization. A well-clustered component (A), with blocking temperatures up to 600°-620°C, is clearly an overprint. A weaker and more dispersed ChRM was isolated above these tem-

TABLE 1. Paleomagnetic Results From the Kirghiz Range

		In Situ				Tilt-Corrected			
L	N	D°	I°	K	a95°	D°	I°	K	a95°
1	22/19	258	-50	20	7.3	235	-46	7	3.8
2	10/9	149	-11	34	8.1	168	-55	34	8.1
3	25/22	141	-37	7	11.5	162	-45	15	7.8
Mean for locality 3A (2+3)									
	35/31	144	-29	8	9.2	163	-48	16	6.1

L is the number of locality, as given in the text; N is number of samples studied/accepted; D and I are declination and inclination, respectively; K is concentration parameter; and a95 is 95% circle of confidence.

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TABLE 2. Results of Fold Test Application

L	C	T	N	G	DF	RA	BC	AC	CV
1		CP	19		36/36	3.7			1.8
		M	19	2	2/34		16.9	0.5	3.3
3		CP	22		42/42	2.15			1.7
		M	22	3	4/38		13.8	0.4	2.6
3A		CP	31		60/60	2.25			1.5
		M	31	4	6/54		14.8	0.9	2.3
4*		CP	44		86/86	6.2			1.4
4	B	CP	39		76/76	11.2			2.7
		M	39	7	12/64		63.5	3.0	2.4 (3.5)
		M**	24	3	4/42		73.4	0.9	2.6
4	A	CP	21		40/40	3.7			1.7
		M	21	3	4/36		124	52.3	2.6

L is locality; C is component (for locality 4 only); T is modifications of the fold test, labeled as in the text; N is number of samples; G is number of groups; DF is number of degrees of freedom; RA is ratio of concentration parameters [McElhinny, 1964]; BC and AC are the calculated F statistics before and after tilt correction, respectively; and CV is the critical value at the 95% (99%) confidence level.

*B component from all sites and data from site 7 are combined.

**The test was performed for sites where the number of hand samples exceeds five.

peratures (component B). Both components are clearly visible on stereonet and orthogonal plots (Figure 4c-4e), although they have rather similar directions and the same polarity. There is only one sample of tuffaceous siltstone (sample 3311, Figure 4c), where component A of normal polarity is antipodal to the reversed site mean; after tilt correction, D = 48°, I = 69° and D = 217°, I = -71°, respectively.

ChRM directions in the dyke are similar to those of sediments, either close to this dyke (site 9) or about 200 m away (site 8). They might have resulted from complete remagnetization of all rocks. On the other hand, both dyke and sediments may be of similar age, and hence they could have acquired their magnetization in the same ancient field. Until more precise ages of rocks are available, the contact test should be considered as inconclusive.

Mean direction for site 7 was found to deviate considerably from the other site means for component B (Figure 5b), although the CP test is positive for the whole set. When the deviating direction is omitted, the CP test is positive again, whereas the M test rejects the null hypothesis after tilt correction at the 95% confidence level and accepts it at the 99% level (Table 2). Thus a prefolding component prevails in the collection, but some distorting factor has affected the data. We do not think it is an unresolved weak secondary component. As many sampled sections are short, paleomagnetic noise, secular variations first of all, could be inadequately averaged within each site. Actually, the M test is positive after tilt corrections for sites with more than five hand samples (Table 2) and the best grouping of paleomagnetic vectors was achieved at 100% unfolding. Thus we think that a prefolding and probably primary component was isolated here.

Site 7 was sampled from the most deformed central part of the fold, where weak cleavage was recognized in the field. When paleomagnetic di-

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rections from this site are combined with those of component A from sites 3 and 9, the three data sets agree much better at an intermediate stage of proportional incremental unfolding (Figure 6a). However, the best clustering of unit vectors can be achieved if differentiated unfolding is applied (Figure 6b and Table 3). This was done by looking for the intersections of small circles drawn through the corresponding in situ and tilt-corrected paleomagnetic directions. It is clearest at site

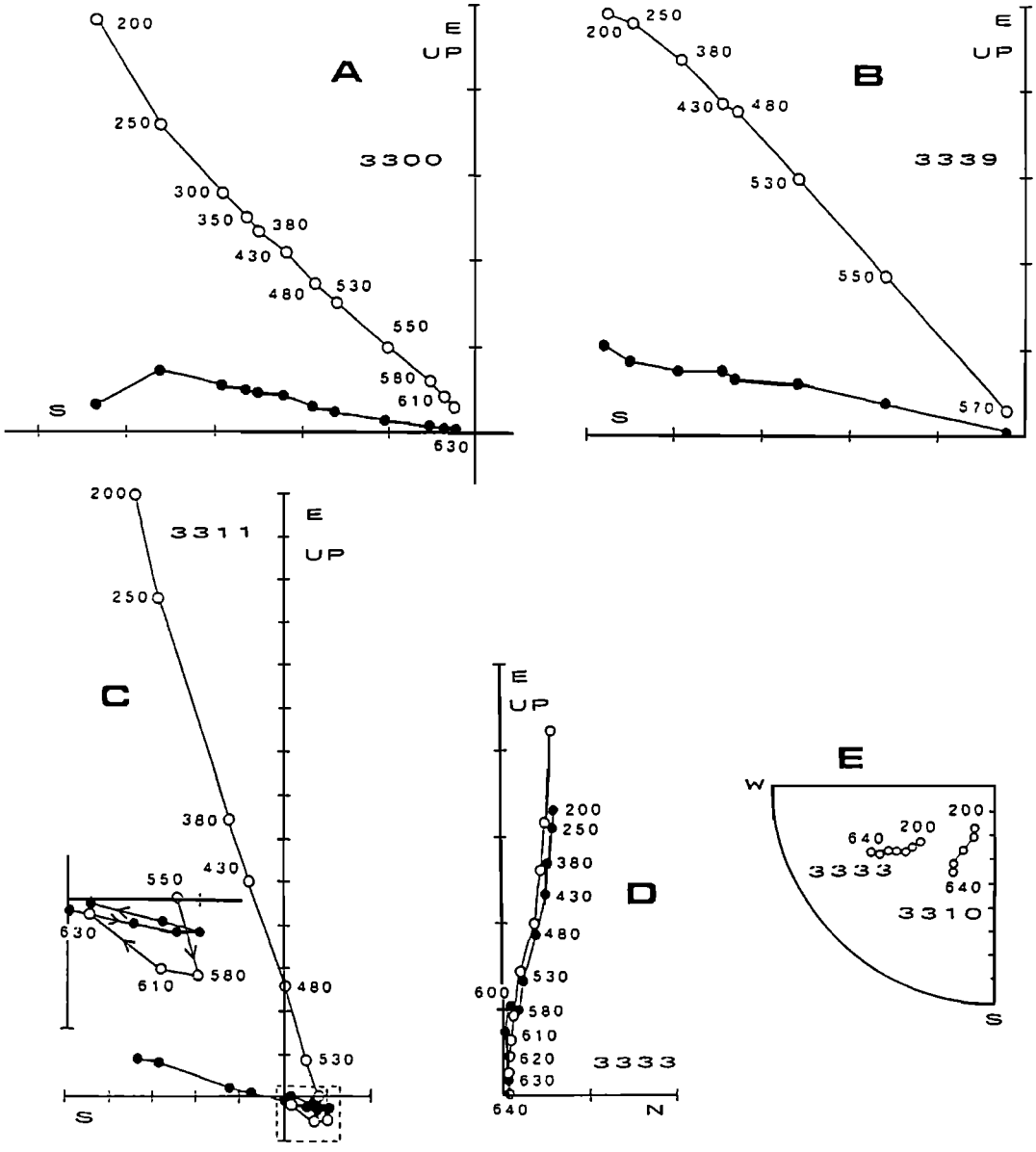


Fig. 4. Representative vector plots of Permian rocks from locality 4: (a) andesite from site 2, (b) diabase from site 9, (c) tuffaceous sandstone from site 3, and (d) red sandstone from site 8. The notation is the same as in Figure 2. Tilt-corrected magnetization directions during progressive thermal demagnetization in Figure 4e are plotted on the upper hemisphere of equal-area projection; sample 3310 is tuffaceous sandstone from site 3 and sample 3333 is red sandstone from site 8. For clarity, not all directions are shown; only the first and last heating steps are labeled.

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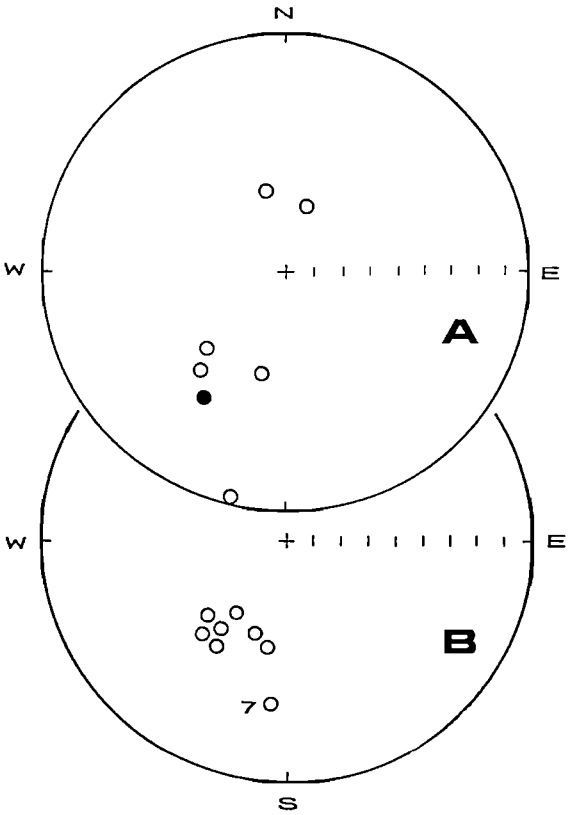


Fig. 5. Site mean directions of ChRM (component B) of Permian rocks from locality 4 (a) before and (b) after tilt correction. The deviating vector from site 7 is shown in stratigraphic coordinates only. The notation is the same as in Figure 3.

mean level, though only three means were used (Table 3). Thus the component A is of synfolding age, but its mean direction depends on the chosen mode of unfolding (Table 3). These rocks were deformed later in the Permian and/or in the Triassic. It is well known that reversed polarity prevailed in the Permian. The Illawara mixed superchron began in the Latest Permian (Tatarian). The presence of the sample with component A of normal polarity suggests a possible Illawara age for this remanence.

The prefolding ChRM that was successfully isolated at four localities has reversed polarity everywhere and was most likely acquired during the Kiaman superchron. Though the conglomerate test is not very convincing, the magnetization of boulders predates the deposition of the intraformational conglomerates. Thus the ChRM can be considered as primary. Three mean vectors (locality 1, 3A and 4) are based on large collections and each direction can be given an independent tectonic interpretation.

INTERPRETATION

The measured inclinations and declinations were compared with the reference directions calculated from an apparent polar wander path (APWP) of the European plate. Values of flattening F and rotation R together with their confidence limits were computed [Demarest, 1983]. Though the European data are rather consistent as indicated by the general agreement of curves derived from different APWPs in Figure 7 [Khramov et al., 1982, Van der Voo, 1990], there are some misfits for the Permian, especially in inclinations. For final analysis, the combined Europe-North America APWP [Van der Voo, 1990] was chosen, as it is based on a larger amount of the up-to-date results.

The rocks studied are rather imprecisely dated as Permian [Masumov et al., 1978]. On plotting the measured inclinations between the Early and Upper Permian reference values, all the data are in acceptable agreement (Figure 7a). The best fit may be achieved if a somewhat older age, e.g., mid-Early Permian, is assumed for these rocks and their remanence. It follows that this part of the Tien Shan did not move with respect to Europe.

Mean inclinations of the synfolding component A (locality 4) are placed arbitrarily at the Permo-Triassic boundary (Figure 7a). The value after proportional unfolding better agrees with the APWP if we assume the Late Permian or older age for this component. This implies an initial stage of folding very close to the rock age. In contrast, the inclination after differentiated unfolding is closer to the Early Triassic reference value. This option is preferred considering the normal polarity component A in one of the samples. It implies an Early Triassic or still younger age for the final stage of deformation of Paleozoic rocks, at least at this locality. More data are clearly needed to confirm or disprove this interpretation.

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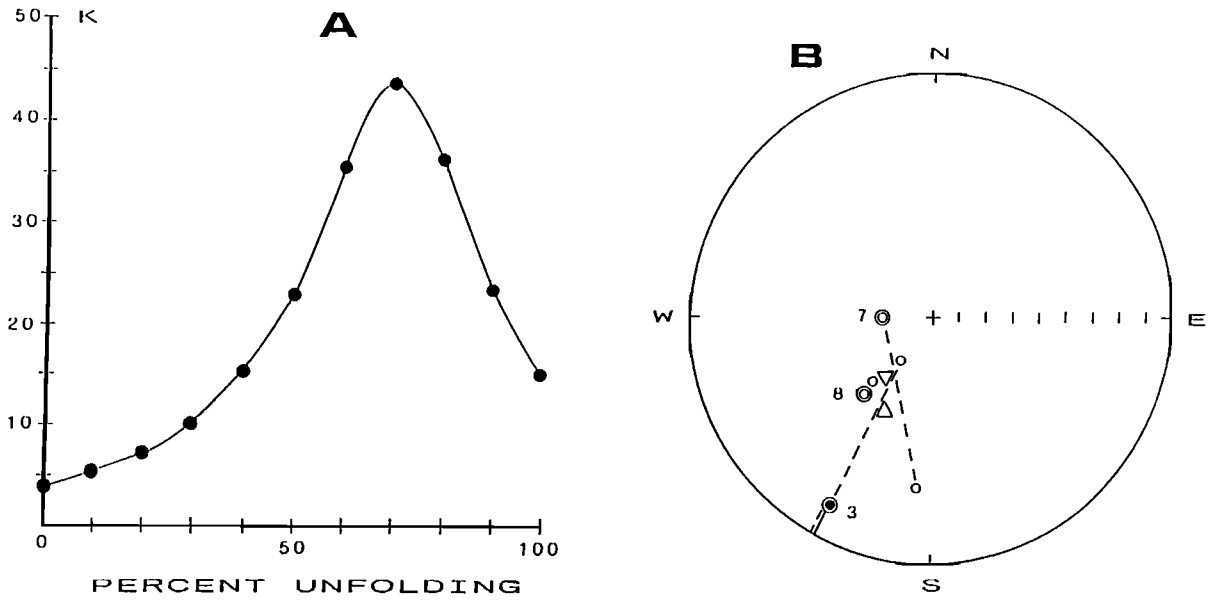


Fig. 6. (a) Precision parameter (K) values versus percent unfolding for component A derived from sites 3, 7, and 8, locality 4. (b) The differentiated unfolding applied to component A mean directions and mean vector for site 7. Single (double) dots denote site mean directions after (before) tilt corrections connected by the corresponding small circle (dashed lines). Triangle (inverted triangle) denotes component A overall mean after proportional (differentiated) unfolding as described in the text. The notation is the same as in Figure 3.

Locality 1 is situated near the northern limit of the Tien Shan, and the corresponding declination agrees well with the reference one (Figure 7b). It can be concluded that the northernmost part of the Tien Shan suffered no relative movement with respect to the European plate. In contrast, mean declinations for localities 3A and 4 are deflected westward, but the very magnitudes of the rotations are different, R = 72°±9° and R = 22°±5°, respectively (the Early Permian reference pole was used). At both localities, folds with nearly horizontal axes were studied, and the fold test points to prefolding age of remanence. Therefore the declination anomalies are stipulated not by plunges of fold axes but by rotations about vertical axes [Bazhenov, 1979]. The declination of the synfolding component recovered from sediments at locality 4 is deflected westward too and the R value (about 20°) is very close to that for component B. Thus the rotation here postdates the initial stage of folding. Note that estimates of rotation are not affected by the age uncertainties (Figure 7b).

As noted earlier, Paleozoic rocks in this area were affected by later deformations dated usually as latest Paleozoic or early Mesozoic. Almost no tectonic activity from the Jurassic through to the mid-Paleogene has been found in the whole Tien Shan. In late Cenozoic, Paleozoic rocks were affected by Alpine deformation. The lowermost member of the Cenozoic succession, the Eocene-Oligocene red beds and basalts, were studied at locality 2, as well as at other localities to the east and south of these area. Paleomagnetic results on these rocks revealed no detectable Cenozoic rotations within the Tien Shan north of the Talas-Fergana fault (J.C. Thomas et al., A paleomagnetic study of Tertiary formations from the Kirghiz Tien Shan and its tectonic implications, submitted to Journal of Geophysical Research, 1992,). It seems reasonable to assume that the area studied had not been affected by Alpine rotations

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TABLE 3. Paleomagnetic Results From Locality 4

			In Situ				Tilt-Corrected			
S	C	N	D°	I°	K	a95°	D°	I°	K	a95°
1	B	4/4	346	-59	29	13.1	212	-57	29	13.1
2	B	5/5	19	-65	30	11.4	189	-50	46	9.3
3	A	9/8	208	12	54	6.7	217	-71	81	5.5
	B	9/7	213	34	34	9.3	214	-48	63	6.7
4	B	3/3	194	-1	306	4.6	197	-52	279	4.8
6	6	4/3	193	-50	35	13.8	213	-44	35	13.8
7	A?	5/5	272	-72	30	11.4	185	-28	63	7.9
8	A	10/8	223	-54	210	3.4	224	-57	176	3.7
	B	10/9	220	-41	40	8.4	220	-45	35	9.0
9	B	10/8	225	-48	32	8.8	225	-49	34	8.5
Mean direction, samples, component B										
		50/39	220	-44	3	16.3	213	-49	33	3.9
Mean direction, seven sites, component B										
			216	-48	<3	-	210	-50	73	6.2
Component A* (70% of unfolding)										
							210	-51	39 (43)	4.9
Component A* (differentiated unfolding)										
							216	-59	66 (185)	3.8

S is site number; C is component of NRM. Numbers in brackets are for concentration parameter for three site means. All other symbols are labeled as in Table 1.

*statistics on the sample level (21 samples in total).

either and declination anomalies at localities 3A and 4 are late Paleozoic-early Mesozoic in age.

Tectonic zones in southern Kazakhstan and the Tien Shan and their boundaries are more or less parallel to each other, and the very zonation is based on the pre-Permian geology of the region [Khain, 1977]. Locality 1 appear to be unrotated at all, while the rotation angle for locality 3A differs considerably from that for locality 4; still further to the south, the magnitude of counterclockwise rotations increases again, from about 60° in the Central Tien Shan north of the Talas-Fergana fault up to 100°-120° south of it [Bazhenov, unpublished data, 1991; Panov, 1991]. If large dimensions of rotated blocks are assumed and the late Paleozoic rotations are taken into account, the pre-Permian tectonic zonation would be greatly distorted, which is unlikely. It is more likely that the observed pattern is due to rotations of small blocks in domino fashion under strike-slip conditions. Judging from the signs of rotations and the approximately E-W strike of major faults, these movements should be sinistral. Except for major faults, the fault network in the Tien Shan is complex, and it is not easy to know which faults are related to the rotations. Besides, if a system of domino blocks rotates through an angle larger than 45°, it should be rearranged at least once [Nur et al., 1986], and no simple fault pattern would be observed. Some rotations have been proposed for the western part of the Kirghiz Range [Kiselev and Korolev, 1964] where the fault network and displacements along separate faults are similar to those in a sinistral strike-slip zone; however, no such data are known so far from the studied area. As for the difference in rotation angles of localities 3A and 4, laboratory modeling has shown that, even under rather simple and uniform strike-slip conditions, the angles of rotations may vary considerably

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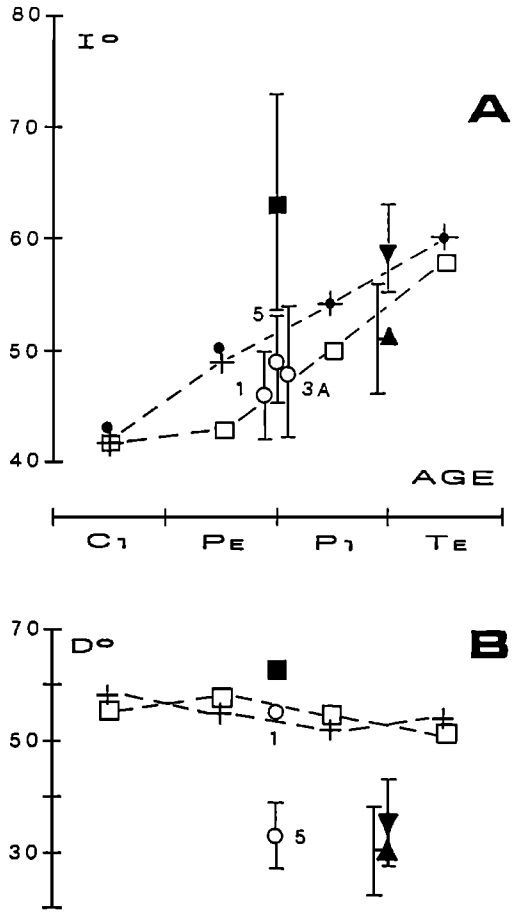


Fig. 7. Plots of (a) inclination and (b) declination versus age for the results from the Northern Tien Shan and reference data. All data are shown as normal polarity directions. Open circles are locality mean values of ChRM (Table 1 and 3), Triangle (inverted triangle) is synfolding component mean direction after proportional (differentiated) unfolding, Crosses are reference data derived from the combined North America-Europe APWP [Van der Voo, 1990]; Solid dots (on inclination plot only) are reference data derived from the European APWP with quality factor, Q > 3 [Van der Voo, 1990]; Open squares are reference data derived from the European APWP [Khramov et al., 1982]. Full square is reference direction derived from the mean Permian pole for Siberia [Khramov et al., 1982]. Vertical lines denote error bars plotted if necessary. The mean declination for locality 3A is not shown because its deviation from the reference data is evident.

from block to block; sometimes, small blocks can even be rotated in an opposite direction [Cobbold et al., 1989]. Such nonuniform rotations seem to be even more probable in an area which had been deformed prior to strike-slip movements. Whatever the mode of rotation, it can be concluded that late Paleozoic and/or Triassic horizontal movements were not restricted to the area close to the Talas-Fergana fault, as deduced from geological data alone [Burtman, 1976].

COMPARISON WITH OTHER REGIONAL STUDIES

Both Lower and Upper Permian rocks have been studied in the Tenghiz basin (about 52.0°N, 67.5°E) in central Kazakhstan [Kumpan et al., 1968]; but the methods and treatments used are not adequate from a modern point of view. Also, Permian (without subdivision) and Early Triassic results were reported from an area in eastern Kazakhstan (about 47.5°N, 80.0°E), although only as entries in catalogues [Khramov, 1986]; thus it is difficult to evaluate their reliability. Nevertheless, it was decided to analyze these data (referred to as CK1, CK2 EK1 and EK2; characters stand for central (CK) and eastern (EK) Kazakhstan and digits for the Early (1) and Late (2) Permian). Recently, presumably Permian data based on a widespread overprint have been reported from west Junggar in China [Zhao et al., 1990].

For brevity, F and R values for central and eastern Kazakhstan and west Junggar as well as the European APWP of Van der Voo [1990] are shown on the same diagram (Figure 8; also see Table 4). F values for Kazakhstan either fall within the error limits or exceed them by 1° at most. Taking into account the reliability of these data, one can conclude that within the error limits, the European platform, Kazakhstan, and the Northern Tien Shan have not suffered any substantial relative poleward motions, at least since the Permian. The F value for Junggar will depend on the assumed overprint age; it is statistically significant for the Late Permian, marginal for the Early Permian, and zero for the Late Carboniferous. In contrast, R values

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change in just the opposite sense (Figure 8 and Table 4). The overprinting remanence was recovered from rocks not younger than Middle-Late Carboniferous, and the authors of the data [Zhao et al., 1990] favor a Permian age for this overprint, their conclusions being based on geological data. If we accept this age, some poleward motion of the area should be assumed as well. However, such a motion is unlikely, since pre-Permian structures can be traced from Junggar to "immobile" eastern Kazakhstan. The Late Carboniferous age implies no poleward movement but some counterclockwise rotation of Junggar. These data are not based on prefolding magnetization and the structural control may be somewhat imperfect. As a result, the mean overprint direction might be biased somewhat by younger tectonic movements. Thus the available data are not sufficient for a definite conclusion on the Late Paleozoic movements of Junggar.

R values for central Kazakhstan are insignificant, while those for eastern Kazakhstan are positive and significant (Figure 8 and Table 4). It may be that they are the result of some clockwise rotation; however, because of the unknown reliability of these data it is not worthwhile discussing the subject more thoroughly.

Finally, we would like to discuss Eurasia as a whole. The European and Siberian plates and Kazakhstan are often thought to have become welded together by the end of the Paleozoic [e.g., Zhao et al., 1990] and to have suffered no noticeable relative horizontal movements since then. When the corresponding late Paleozoic mean poles are plotted together

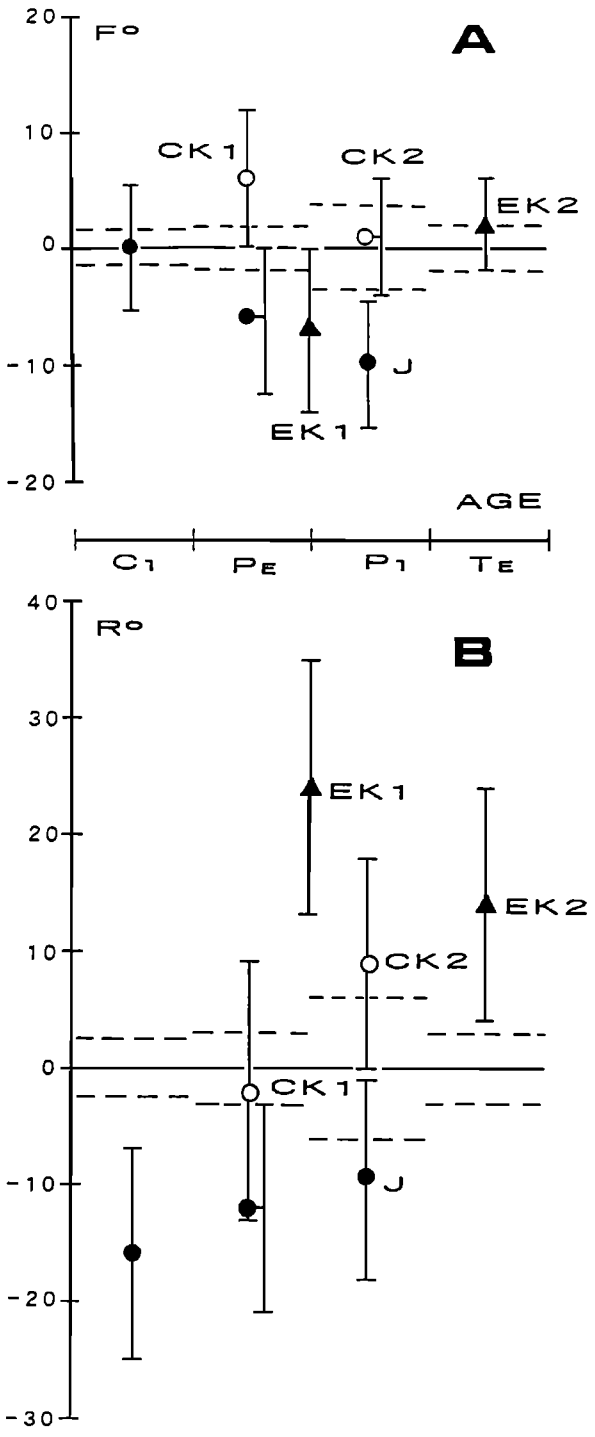


Fig. 8. Plots of (a) flattening F and (b) rotation R versus age for the results from Kazakhstan and Junggar: CK1 and CK2 (open dots), the Early and Late Permian data, respectively, from central Kazakhstan [Kumpan et al, 1968]; EK1 and EK2 (triangles), the Permian and Early Triassic data, respectively, from eastern Kazakhstan [Khramov, 1986]; J (solid dots), the data from Junggar [Zhao et al., 1990]. The latter are plotted in different positions with respect to the age axis as discussed in the text. Thin solid vertical lines denote error bars for each result; dashed horizontal lines represent error bars associated with mean poles from the combined North America-Europe APWP [Van der Voo, 1990].

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TABLE 4. Kinematic Parameters for Kazakhstan and Junggar With Respect to the European and Siberian Reference Data

			Europe						Siberia*		
L	I_m°	D_m°	I_r°	F°	dF°	D_r°	R°	dR°	I_r°	F°	dF°
CK1	56	53	50	6	5	55	2	9	62	-6	8
CK2	56	45	50	1	6	53	-8	10	62	-6	7
EK1	50	82	57	-7	6	60	22	10	66	-16	8
EK2	66	74	64	2	4	58	16	9			
J_{P1}	50	48	60	-10	6	57	-9	10			
J_{Pe}			56	-6	5	60	-12	7			
J_{C1}			50	0	4	64	-16	7			

L is locality and age of data as described in the text; D and I are declination and inclination (converted to normal polarity); F is flattening; dF is semi-interval of confidence; R is rotation angle (positive and negative values stand for clockwise and anticlockwise rotations, respectively); dR is semi-interval of confidence. Subscripts "m" and "r" denote measured and reference data, respectively. Three entries for data from Junggar correspond to different ages of the remanence as discussed in the text.

*F values only are calculated because predicted declinations are very similar to the European data.

on a stereonet [Zhao et al., 1990], the agreement seems to be good. At the same time, a systematic discrepancy exists between the Latest Permian and Early Triassic poles of these two plates. This discrepancy has been explained by relative movement of the two plates due to opening of the West Siberian basin, which ceased before the Late Jurassic or Early Cretaceous [Khramov et al., 1982; Bazhenov and Mossakovsky, 1986]. However, a rather severe data selection was applied to the European and Siberian data sets to evaluate this discrepancy numerically [Bazhenov and Mossakovsky, 1986]. Of two mean Early Triassic poles cited there, the European one deviates considerably, both from the Siberian pole and from the mean pole of Van der Voo [1990], although the latter two are in close agreement. Therefore this discrepancy may be an artefact due to data selection. Siberian Permian poles are scattered (K = 19), and the corresponding mean pole at 45°N, 141°E is poorly defined (A_{95} = 10°) [Khramov et al., 1982]. Nevertheless, it differs considerably from the European ones. As the result, the Siberian mean reference inclination is deflected from the measured ones in the Northern Tien Shan (Figure 7a). For the three Permian results from Kazakhstan (Table 4), the F value is almost equal to error limits in two cases but is clearly significant in the third case. Therefore no definite conclusion can yet be drawn. It should be noted that the main part of Siberian data was obtained some 15 or 20 years ago and its reliability is now debatable. It may be that all the discrepancies will disappear in the future, when more reliable results become available for Siberia, and the hypothesis on the relative movement of these two platforms will be tested more rigorously.

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